CEREBRO Solution Architecture

Prepared by Joselito Dizon

Containers, data stores and the pipeline from upload to an answer with citations.

Context

This architecture was baselined on 2 August 2026. It turns the vision note into a buildable system and applies the patterns from the lectures on Knowledge Graphs and Retrieval-Augmented Generation.

Containers

The web tier is Next.js; the API is FastAPI; PostgreSQL holds users, artefacts, fragments and their vectors. The browser only calls the web origin, which proxies the API, so the session cookie stays first-party.

Pipeline

An artefact moves through upload, inspection, extraction, review and approval. Approval re-verifies the checksum, splits the text into fragments and embeds them for Semantic Search. Each fragment records its Provenance: document, page and character range.

Recall retrieves passages first and lets a local model served by Ollama phrase the answer, with citations.

Deployment

Containerised Deployment targets Kubernetes in a cerebro namespace, with Docker Compose as the simpler path. Host ports are configurable because a shared machine often has its usual ports taken.

Illustrative sample document for the CEREBRO demo library.